

David SERCK

Full-Stack Developer

Specialized in Angular, Python, and Java/Spring ecosystems.
Experienced in building APIs, business workflows, and user-centric interfaces.

Available immediately.

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Experience

Konekto
Ottignies, Belgium

Developer
Mar 2024 - Apr 2026

Processing of electronic invoices (PEPPOL/UBL) and procurement workflows.

HR time-tracking system and automated integration with social payroll providers.

Technical migrations to Java 21, Spring Boot 4.0.1, and Angular 21.

Schedule optimization via AI (ChatGPT) and API monitoring dashboards.

Custom authentication and permission management via Keycloak.
https://konekto.be

Dstny (telecom company)
Wavre, Belgium

Cloud developer
Feb 2021- Feb 2024

Lookup server for telephone numbers.

APIs in python for managing subscribers and pbx settings.

Several websites for end user settings in react and angular.
https://www.dstny.be

Micro Belgium Application (fiduciary)
Wavre, Belgium

Junior developer
Nov 2017 - Jul 2020

Import of Belgian electronic invoice (e-fff) directly into accounting software.

Synchronization of databases with google classroom for a school.

PDF rendering of Belgian coded account statement (CODA).

Calendar for an orthopaedic firm.

Improvement of existing software “HappyPhone” (a solution in C# for logging phone calls in a google spreadsheet).

Education

IFOSUP, Wavre
Bachelor in business informatics
2014-2018

(“Bachelier en informatique de gestion”) (great distinction)

Technobel, Nivelles
Front-end developer training
Sep 2020 - Dec 2020

Database, python, javascript, html, css, angular, node js, react and scrum.

Technofutur, Louvain-La-Neuve
Java Full-Stack
Jun 2024 - Aug 2024

Technofutur TIC, Charleroi
Architecte IA/IOT
Jul 2026 - Present

Skills

Backend & Architecture
Java 21, Spring Boot (2.x/3.x), JPA, Maven/Gradle, Python (FastAPI / Tornado), C# (.NET Core)

Frontend
Angular (19 to 21), React, TypeScript, MobX, Flutter/Dart, Material Design, Ant Design

DevOps, Cloud & Security
Ansible, GitLab CI/CD, Keycloak (IAM), Kong API Gateway, Docker Compose

Databases
T-SQL, Redis, CockroachDB, MariaDB, PostgreSQL, MySQL, H2

AI
YOLO computer vision, OpenAI API

Languages

French	English	Spanish	Chinese
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Projects at Konekto

Digiprocs/Euroports

Context:
Improvement and maintenance of a B2B purchase order and invoicing management system.

Achievements:

Development of features for processing invoices in PEPPOL format, management of validation workflows, and updates to Spring Boot and Angular.

Tasks:

- Processing invoices/credit notes from the PEPPOL network (UBL format).
- Creation of a page with pagination and search filters to list PEPPOL documents awaiting completion or validation.
- Implementation of an invoice validation system by department/service, using Keycloak attributes to manage permissions.
- Management of decimal quantities and negative amounts on purchase and delivery orders.
- Modification of the business logic to link multiple documents to an invoice/order, to unlink an invoice from a purchase order, and to modify a purchase order even after receiving an invoice.
- Dynamic filters for journal numbers to ensure they all begin with the same number in a purchase order.
- Application migration to Angular 21 (from version 19) and Spring Boot 4.0.1 (from version 3.5.x).

Technical environment and tools:

Angular 21, Java, Spring Boot 4.0.1, Keycloak, PEPPOL (UBL).
https://konekto.be/projets/euroports/

Spurado

Background:
Enhancement of an HR management application with features for tracking working hours and managing leave.

Achievements:

Implementation of a comprehensive "time clock" system for employees, with integration for the social security office, and improvement of the existing leave management system.

Tasks:

- Development of a clocking-in widget allowing employees to record the start and end of their working day as well as their breaks.
- Creation of a calendar page to allow employees to view their hours and administrators to modify them.
- Creation of a secure endpoint dedicated to the social security office for the automated import of hours worked.
- Implementation of a leave classification system with an approval process and a calendar page to view leave.
- Improvement of the layout of PDF reports generated via iText and the associated statistics.

Technical environment and tools:

Java, Spring, Angular, iText.
https://konekto.be/projets/spurado/

Vision

Context:
Modernization of a survey and questionnaire management tool.

Achievements:
Major Java and Spring Boot migrations, addition of user customization features, and improved responsive design.

Tasks:

- Development of a style customization feature for questions and question groups.
- Addition of dynamic filters when exporting survey data.
- Complete technical migration of the backend environment: from Gradle to Maven, from Java 11 to Java 21, and from Spring Boot 2 to Spring Boot 3.
- Application stabilization (fixing numerous bugs such as circular dependencies in Groovy formulas and synchronization errors).

Technical Environment and Tools:
Java, Spring Boot, Maven.
https://konekto.be/projets/vision-360/

Event Locator/AtlasControl

Context:
Modernization of a technical intervention management application and addition of a prompt for artificial intelligence integration to optimize scheduling.

Achievements:
Data migration to the cloud, upgrade of Spring Boot and Angular, and use of the OpenAI API (ChatGPT) for scheduling assistance.

Tasks:

- Migration of attachments stored in a MariaDB database to Amazon S3 cloud storage.
- Development of a context menu querying ChatGPT to obtain recommendations for optimizing the technicians' daily schedules.
- Upgrade of the application to Angular 20 and Spring Boot 3.5.x.

Technical Environment and Tools:
Java, Spring, Angular, MariaDB, Amazon S3, OpenAI API.
https://konekto.be/projets/event-locator/

Loopery

Context:
Retail automation solution for a second-hand children's clothing store, aimed at significantly speeding up product cataloging and inventory processing.

Achievements:

Integration of AI-based computer vision features into an existing mobile application to automate product photography workflows, and enhancement of inventory categorization on the web platform.

Tasks:

Design and deployment of a FastAPI backend integrated with a YOLO object detection model to automatically detect garments and calculate precise image cropping areas.

Modification of the existing Flutter mobile application to display automated crop suggestions, streamlining the item listing process for store staff.

Implementation of intuitive garment categorization features on the Angular web platform.

Technical Environment and Tools:

Python, FastAPI, YOLO (Object Detection), Flutter, Dart, Angular
https://konekto.be/projets/loopery/

Monikonekto

Context:
Need for an internal monitoring tool to visualize the proper functioning of the company's application ecosystem.

Achievements:

Creation of a simple monitoring dashboard.

Tasks:

Development of a web page centralizing the visualization of the status (HTTP status) of the various Konekto applications and APIs.

Implementation of a system for retrieving and displaying detailed JSON error messages, thus facilitating debugging for the technical team.

Technical Environment and Tools:

Vanilla JavaScript, HTML, CSS

Login page for keycloak

2024 / 2 weeks

Context:
Konekto wanted to enhance their keycloak login page for one of their client and for themself.

Achievement:

I designed several login pages composed of one part with an illustration and an animated background and another part with a form. Between the two parts there is a transition with an svg mask.

Tasks:

- Designed several login pages in freemarker format.
- Changed the Java Spring Boot backend to send emails using the same freemarker template.

Technical Environment and Tools:

I used a docker-compose and mailhog to test the login page.

Projects at Dstny

Admin interface for external communication system

2023-2024

Context :

The “Communication System Front End Administration” is a web interface and an API that allow (a tenant) to connect subscribers to external communication platforms (msteams, mobile, zoom, sip provider ...) in addition to the landline.

Achievement :

- Created the API + tests (from scratch)
- Helped on others API for specific platforms
- Parts of the React frontend (form, search)
- Parts of the React frontend and backend for the debugging tool to simulate call routing.

Technical Environment and Tools :

The CSFE admin is in python3 with the tornado web framework.

Frontend use React.

Use pytest, docker and gitlab CI

Use OpenApi for documentation.

Use the azure cloud for the staging environment.

Web app for phone user settings

2023 / 3 months

Context:

FMPA manage phone subscribers settings for our ‘FMU’ solution

Achievement:

- Diverse work on this API:

o Switched database to postgresql

o Added metrics

o Improved logging

o New endpoints to handle new params and new use cases

o Handled compatibility with multiple external API version

Tasks:

- New releases
- Deployment with ansible playbook

Technical Environment and Tools:

FMPA is in python3 with tornado webframework.

It uses the marsmallow for json schema parsing and openAPI for documentation.

Tracing with opentelemetry.

It uses pytest, docker for tests as well as VCR to cache answer for external APIs.

Phone number lookup server

2022 / 5 months

Context:

This API stores all phone numbers and the respective domain where there are registered.

It also caches sip headers for call in progress.

Achievement:

I rewrote the API along with my colleague and deployed to production.

Tasks:

- Rewrote the API based on the specification of the previous one.
- Made the documentation
- Wrote the tests and created a docker compose to run the tests in gitlab CI.
- Wrote a migration script
- Adapted the ansible playbook to install the new version of the API.

Technical Environment and Tools:

The lookup server is in python3 with the tornado web framework.

Communication to asterisk with AMI + ARI.

Used cockroach database + Redis cache.

API to manage clusters, tenants, subscribers, pbx and calendars settings

2021 / x months

Context:

FMPA manage phone subscribers settings for our 'FMU' solution

Achievement:

- Diverse work on this API:

o Switched database to postgresql

o Added metrics

o Improved logging

o New endpoints to handle new params and new use cases

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Technical Environment and Tools:

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Projects at MBA

Insightly contacts synchronization

2020 / 2 months

Context:

IAB is the European-level association for the digital marketing and advertising ecosystem. They use the platform “Insightly” to manage their contacts. I worked alone to create a synchronization between this platform, Google Groups and Google Sheets.

Achievement:

- Synchronization between this platform and google Groups
- Synchronization between this platform and google Sheet

Tasks:

- Because it is not possible of retrieving the lists of contacts with the Insightly API a configuration file was used to store the search criteria for building those lists. Using this configuration file, several queries to the API were combined to filter the contacts.
- The synchronization was executed periodically.
- At first, the synchronization with a list of spreadsheets was done in the same script as the one for google groups but as the number of lists demanded increased a second script was used for a better separation of the process.
- Google spreadsheet API was handled with pygsheets to which I contributed with 2 smalls pull request.

Technical Environment and Tools:

Python was used for those scripts along with Windows task scheduler.

Import of e-fff (Belgian electronic invoice) directly into Winbooks (accounting software)

2020 / 2 months

Context:

My company received from Winbooks a bunch of electronic invoices in the form of e-fff files with .xlsx spreadsheets every month.

Achievement:

- I recovered those files from the mailbox (gmail), parsed them and build a system to import them on demand into the accounting software.
- While importing those invoices some others were generated in order to bill our customers with a reseller margin.
- I set up a page on the intranet to configure the relation between winbooks licences and our customers

Tasks:

- A first script was created to recover every emails matching some (google) regex from gmail along with attachments.
- A second script was run to parse and import invoices into a database.
- Finally, an SQL procedure was run on demand (through a menu in Winbooks logistic software that my colleague made) to generate the invoices to be imported in the accounting software.

Technical Environment and Tools:

Python was used for those scripts along with windows task scheduler and Sql Server Express Database 2016.

T-sql script to create synchronization procedures between Logistics .dbf files and sql server

2020 / 1 months

Context:

Micro Belgium Application work with the desktop (classical) version of Winbooks and Logistics which did not use a database but a bunch of file with a .dbf extension. Our company automated a lot of operations using Sql procedures but reading .dbf files with sql “openrowset” command was painful. I worked alone in order to create a generic SQL procedure to synchronize the content of the relevant .dbf files into a database.

Achievement:

I made this generic SQL procedure which is used for every new client the company acquire.

Tasks:

- I analyzed existing procedures for several of our big clients.
- I made a first Sql procedure to create a linked server for a specific Winbooks folder
- Then I made a second Sql procedure using the first one which and given some parameters generated a database along with a synchronization procedure for the given client.

Technical Environment and Tools:

I used sql server 2016 procedures

Synchronization of databases with google classroom for a school 2020 / 3 months

Context:

The school had information about students and professors scattered in multiples databases (SQL server, InterBase, Access, MySQL), some rather badly designed. They wanted to use google classroom.

Achievement :

- Each year new students/professors get a google account and are added to the adequate classes on google classroom, google groups, drupal and the website to rent material.

Tasks:

- A python script firstly merged the users in a SQL server database then they were created as google users (with a unique and easy email) and pushed in the google groups and classrooms matching the student’s options.
- Formers professors and students are kept in specials group with the hope they can one day be reached to help promote the school.
- Classes are also synchronized in the website for audio-visual equipment reservation and the school intranet (Drupal) with the perk of having the same password has the one randomly created for google
- (I translated the PHP password hashing function in T-SQL).
- Password is sent with a python script.
- The following school year, I made an additional script to email all students that never connected to their google account with a new password.

Technical Environment and Tools:

Python was used for those scripts along with a sql server (express 2019) database and sql server procedures.

PDF rendering of Belgian coded account statement (CODA)

2019 / 2 months

Context:

Electronic account statement are handy to import in the accounting software but clients missed an user-friendly presentation the operations. I worked alone to generate comprehensive .pdf based on CODA files.

Achievement:

- Pdf are generated every morning for our clients.

Tasks:

- I used a free and open-source external tool (wkhtmltopdf) to quickly render the generated HTML code to PDF
- I made detailed PDF with pagination and information coming straight from the accounting software (e.g. Client ID and detail from card payment)

Technical Environment and Tools:

Python was used for those scripts along with windows task scheduler.

On demand audio transcription of phone calls recording

2019 / 1 months

Context:

Sometime my colleagues listen their previous conversation on the phone to remind themselves of some information they missed. However, those recording can last long. I worked alone to transcribe those audio file on demand.

Achievement:

- A page in the intranet allow the user to request a transcription of a specific phone call recording
- Transcript of the conversation is uploaded to google drive

Tasks:

- I used google Speech-to-Text and google natural language to transcribe audio recording.
- I made a page on the intranet to request the transcription of a specific recording.
- I modified our software for logging calls (HappyPhone) so the google spreadsheet would link to the webpage on the intranet for requesting a transcription.

Technical Environment and Tools:

I used c# .net core mvc 3.1 for this project.

Intranet

2018 / 2 months

Context:

Multiples tools and services were provided on the network but were not used because the team did not know of them. I worked alone to gather everything on an intranet.

Achievement:

- The website allows authorized persons to configure internal processes and access all tools.

Tasks:

- Statistics with google charts
- Interface for configuration of internal processes (CODA, EFFF)
- Interface for quick operation on coda files, efff files, excel files.
- Shortcut to internal tools and resources
- Authentication (with google account), authorizations by roles
- Translations with .po files.
- REST endpoint for phone terminals

Technical Environment and Tools:

.Net MVC Core 3.1 + Jenkins